

Week 1 Synthesis: LLM Tool Competencies & Failure Modes

Perplexity

Strengths: Fast web search, cites sources, good for current events and factual summaries. **Failure patterns:** Sources don't always support claims; shallow analysis; mixes facts with assumptions; confident hallucinations. **When NOT to trust:** Complex reasoning, nuanced topics, critical decisions, unverified sources, anything requiring depth beyond surface facts.

Claude

Strengths: Shows reasoning; admits uncertainty; handles context well; strong on technical analysis; transparent about limitations. **Failure patterns:** Knowledge cutoff (Jan 2025); over-explains simple questions; can construct plausible-sounding false details; reasoning $\neq$ accuracy. **When NOT to trust:** Current events/news; speed-critical queries; unverified facts; recent policy/API changes; when you need a quick answer, not a thoughtful one.

ChatGPT

Strengths: Balanced speed/depth; excellent writing quality; good instruction-following; natural, readable explanations. **Failure patterns:** Confident without verification; shallow reasoning on complex topics; no source verification; outdated knowledge without signaling it. **When NOT to trust:** Factual claims without sources; technical architecture; recent developments; critical analysis; very long contexts; anything where tone sounds authoritative but sources are absent.

Decision Rules

Need	Use
Current facts	Perplexity (verify sources)
Technical reasoning	Claude (verify critical claims)
Readable explanation	ChatGPT
Unsure	Use two tools; disagreement = dig deeper

Critical Insights

1. **Confidence $\neq$ Accuracy.** The polished answer is often the most dangerous. All three confidently make mistakes.

2. **No tool is trustworthy in isolation.** Fast tools are less accurate; thoughtful tools are slower. Use them to verify each other.
 3. **Context degradation is real.** Long documents, long conversations, edge cases—all three break. Shorter, specific queries get better answers.
 4. **Source attribution matters.** Perplexity's citations aren't guarantees but are a reliability signal. Lack of sources means verification is your job.
 5. **Failure modes are distinct.** Perplexity hallucinates facts; Claude over-caveats; ChatGPT sounds certain without evidence. Know which failure to expect.
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What NOT to Trust: Quick Checklist

- ☐ Current information without Perplexity verification?
- ☐ Critical decision based on one tool?
- ☐ Confident claims without sources?
- ☐ Very long context (expect degradation)?
- ☐ Edge cases untested?

Rule of thumb: If the answer matters, use two tools. If you only have one, assume it's wrong about something.